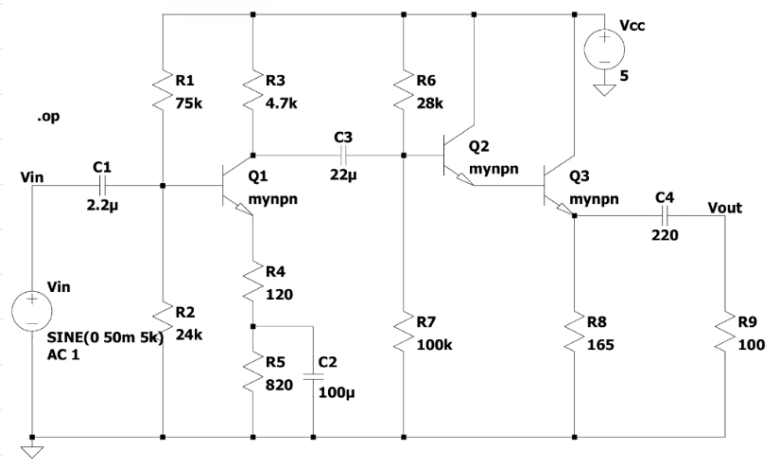
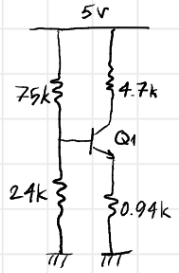
سها نیرومند. 403170569. پروژه الکترونیک 1. استاد فخارزاده

.1

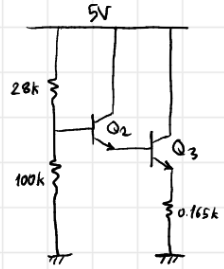




$$V_{B1} = \frac{24}{99} \times 5 \simeq 1.21v \qquad V_{E1} = 1.21 - 0.7 = 0.51$$

$$I_{E1} = \frac{0.51}{0.94} \simeq 0.54\ mA \simeq I_{C1}$$

$$-5 + 4.7 \times 0.54 + V_{CE1} + 0.94 \times 0.54 = 0 \longrightarrow V_{CE1} \simeq 1.95v$$



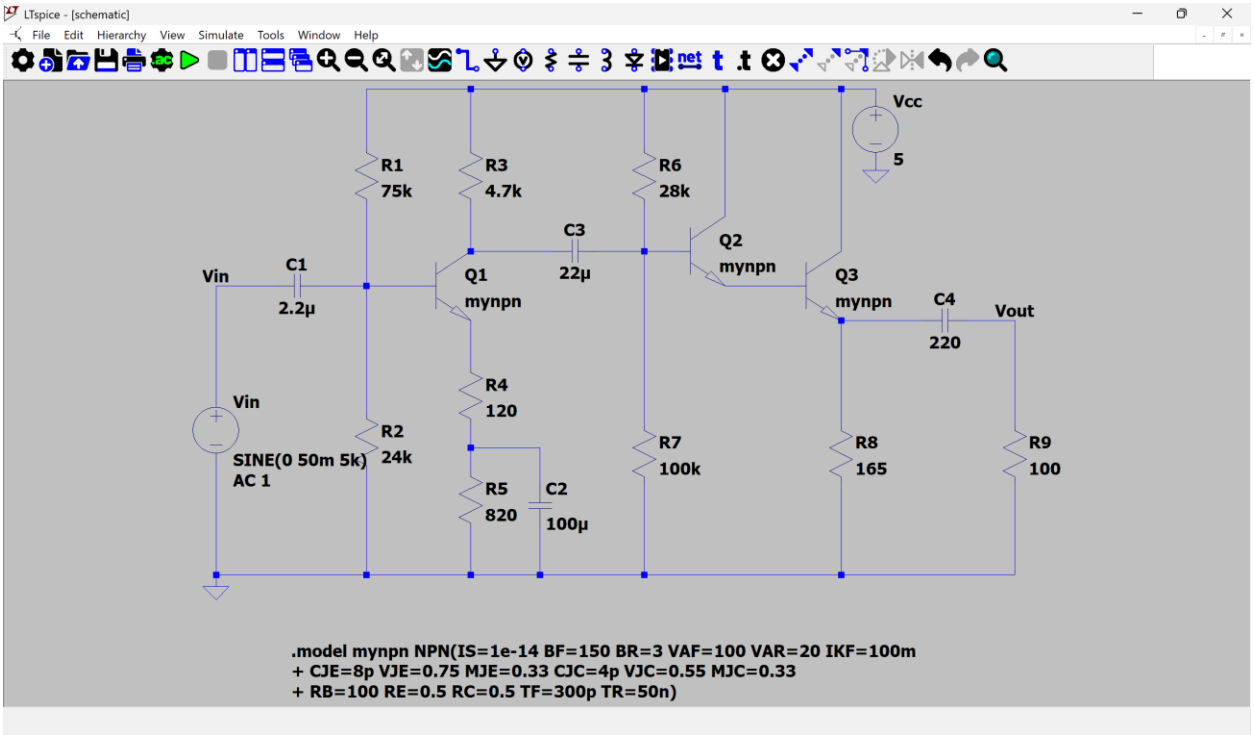
$$V_{B2} = \frac{100}{128} \times 5 \simeq 3.91 \qquad V_{E1} = V_{B2} = 3.91 - 0.7 = 3.21 \qquad V_{E3} = 3.21 - 0.7 = 2.51$$

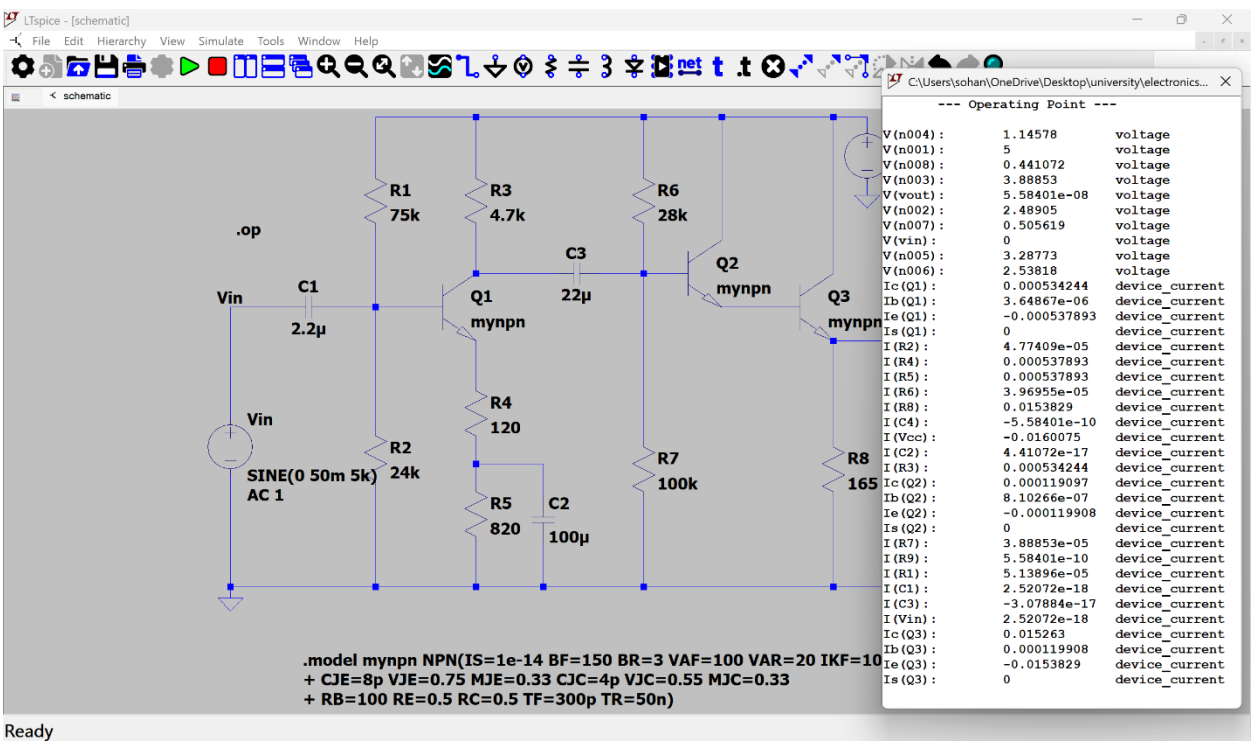
$$I_{E3} = \frac{2.51}{0.165} \simeq 15.21mA = 151\, I_{B3} \longrightarrow I_{B3} = 0.1mA = I_{E2} = I_{C2}$$

$$I_{C3} = 15.11mA$$

$$-5 + V_{CE3} + 0.165 \times 15.21 = 0 \longrightarrow V_{CE3} \simeq 2.49v$$

$$-5 + V_{CE2} + 0.7 + 0.165 \times 15.21 = 0 \longrightarrow V_{CE2} \simeq 1.79v$$


LTspice - [schematic]
File Edit Hierarchy View Simulate Tools Window Help
Vcc
5
R1
75k
R3
4.7k
R6
28k
C3
22µ
Q2
mynpn
Vin
C1
2.2µ
Q1
mynpn
Q3
mynpn
C4
220
Vout
R4
120
Vin
R2
24k
SINE(0 50m 5k)
AC 1
R7
100k
R8
165
R9
100
R5
820
C2
100µ
.model mynpn NPN(IS=1e-14 BF=150 BR=3 VAF=100 VAR=20 IKF=100m
+ CJE=8p VJE=0.75 MJE=0.33 CJC=4p VJC=0.55 MJC=0.33
+ RB=100 RE=0.5 RC=0.5 TF=300p TR=50n)


LTspice - [schematic]
File Edit Hierarchy View Simulate Tools Window Help
schematic
.op
R1
75k
R3
4.7k
R6
28k
C3
22µ
Q2
mynpn
Vin
C1
2.2µ
Q1
mynpn
Q3
mynpn
R4
120
Vin
R2
24k
SINE(0 50m 5k)
AC 1
R7
100k
R8
165
R5
820
C2
100µ
.model mynpn NPN(IS=1e-14 BF=150 BR=3 VAF=100 VAR=20 IKF=10
+ CJE=8p VJE=0.75 MJE=0.33 CJC=4p VJC=0.55 MJC=0.33
+ RB=100 RE=0.5 RC=0.5 TF=300p TR=50n)
Ready
C:\Users\sohan\OneDrive\Desktop\university\electronics...
--- Operating Point ---
V(n004): 1.14578 voltage
V(n001): 5 voltage
V(n008): 0.441072 voltage
V(n003): 3.88853 voltage
V(vout): 5.58401e-08 voltage
V(n002): 2.48905 voltage
V(n007): 0.505619 voltage
V(vin): 0 voltage
V(n005): 3.28773 voltage
V(n006): 2.53818 voltage
Ic(Q1): 0.000534244 device_current
Ib(Q1): 3.64867e-06 device_current
Ie(Q1): -0.000537893 device_current
Is(Q1): 0 device_current
I(R2): 4.77409e-05 device_current
I(R4): 0.000537893 device_current
I(R5): 0.000537893 device_current
I(R6): 3.96955e-05 device_current
I(R8): 0.0153829 device_current
I(C4): -5.58401e-10 device_current
I(Vcc): -0.0160075 device_current
I(C2): 4.41072e-17 device_current
I(R3): 0.000534244 device_current
Ic(Q2): 0.000119097 device_current
Ib(Q2): 8.10266e-07 device_current
Ie(Q2): -0.000119908 device_current
Is(Q2): 0 device_current
I(R7): 3.88853e-05 device_current
I(R9): 5.58401e-10 device_current
I(R1): 5.13896e-05 device_current
I(C1): 2.52072e-18 device_current
I(C3): -3.07884e-17 device_current
I(Vin): 2.52072e-18 device_current
Ic(Q3): 0.015263 device_current
Ib(Q3): 0.000119908 device_current
Ie(Q3): -0.0153829 device_current
Is(Q3): 0 device_current

Q1

پارامتر	محاسبه دستی	نتیجه شبیه‌سازی	درصد خطا
V_{CE} (V)	1.95	1.98	1.71%
I_C (mA)	0.54	0.53	1.07%
V_{BE} (V)	0.70	0.64	8.55%

Q2

پارامتر	محاسبه دستی	نتیجه شبیه‌سازی	درصد خطا
V_{CE} (V)	1.79	1.71	4.34%
I_C (mA)	0.10	0.12	19.10%
V_{BE} (V)	0.70	0.62	11.64%

Q3

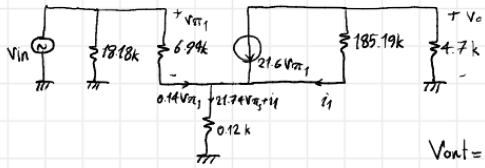
پارامتر	محاسبه دستی	نتیجه شبیه‌سازی	درصد خطا
V_{CE} (V)	2.49	2.46	1.13%
I_C (mA)	15.11	15.26	1.01%
V_{BE} (V)	0.70	0.75	7.08%

$g_{m1} = \frac{0.54}{25} = 21.6\,m\Omega^{-1}$ $\quad r_{\pi_1} = \frac{150}{21.6} \simeq 6.94\,k\Omega$ $\quad r_{o1} = \frac{100}{0.54} \simeq 185.19\,k\Omega$

$g_{m2} = \frac{0.1}{25} = 4\,m\Omega^{-1}$ $\quad r_{\pi_2} = \frac{150}{4} = 37.5\,k\Omega$ $\quad r_{o2} = \frac{100}{0.1} = 1000\,k\Omega$

$g_{m3} = \frac{15.11}{25} = 604.4\,m\Omega^{-1}$ $\quad r_{\pi_3} = \frac{150}{604.4} \simeq 0.25\,k\Omega$ $\quad r_{o3} = \frac{100}{15.11} \simeq 6.62\,k\Omega$

: Av

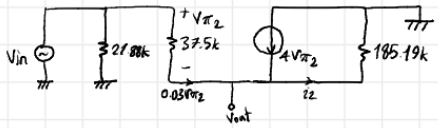


$4.7(21.6v_{\pi_1}+i_1) + 185.19\,i_1 + 0.12(21.74v_{\pi_1}+i_1) = 0$

$\rightarrow i_1 = -0.55v_{\pi_1}$

$V_{out} = -4.7(21.6v_{\pi_1} - 0.55v_{\pi_1}) \simeq -98.94\,v_{\pi_1}$

$V_{in} = v_{\pi_1} + 0.12(21.74v_{\pi_1} - 0.55v_{\pi_1}) \simeq 3.54\,v_{\pi_1}$ $\quad A_{v1} = \frac{-98.94v_{\pi_1}}{3.54v_{\pi_1}} \simeq -27.95$

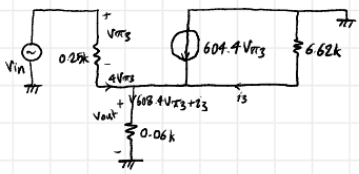


$i_2 = 4.03v_{\pi_2}$

$V_{out} = 185.19\,i_2 = 185.19 \times 4.03\,v_{\pi_2} = 746.32\,v_{\pi_2}$

$V_{in} = v_{\pi_2} + 746.32\,v_{\pi_2} = 747.32\,v_{\pi_2}$

$A_{v2} = \frac{746.32\,v_{\pi_2}}{747.32\,v_{\pi_2}} \simeq 0.99$



$6.62\,i_3 + 0.06(608.4v_{\pi_3} + i_3) = 0 \rightarrow i_3 = -5.47\,v_{\pi_3}$

$V_{out} = 0.06(608.4v_{\pi_3} - 5.47v_{\pi_3}) = 36.18v_{\pi_3}$

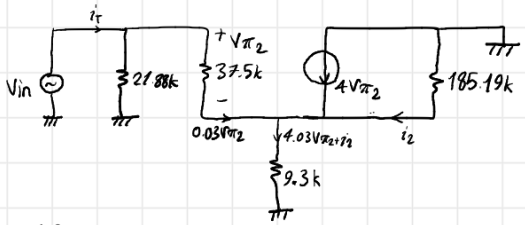
$V_{in} = v_{\pi_3} + 36.18v_{\pi_3} = 37.18v_{\pi_3}$

$A_{v3} = \frac{36.18v_{\pi_3}}{37.18v_{\pi}} \simeq 0.97$

$A_{vT} = -27.95 \times 0.99 \times 0.97 \simeq -26.84$

: Rin

$R_{in3} = \frac{37.18v_{\pi_3}}{4v_{\pi_3}} \simeq 9.3\,k\Omega$



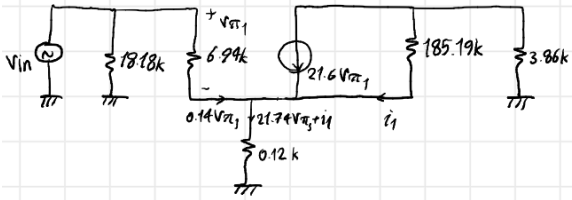
$185.19\,i_2 + 9.3(4.03v_{\pi_2} + i_2) = 0 \rightarrow i_2 = -0.19v_{\pi_2}$

$4.03v_{\pi_2} + i_2 = 3.84v_{\pi_2}$

$V_{in} = v_{\pi_2} + 9.3 \times 3.84v_{\pi_2} = 36.71v_{\pi_2}$

$i_T = 0.03v_{\pi_2} + \frac{36.71v_{\pi_2}}{21.88} \simeq 1.7v_{\pi_2}$ $\quad R_{in2} = \frac{36.71v_{\pi_2}}{1.7v_{\pi_2}} \simeq 21.59\,k\Omega$

$4.7 \parallel 21.59 \simeq 3.86\,k\Omega$



$3.86(21.6v_{\pi_1} + i_1) + 185.19\,i_1 + 0.12(21.74v_{\pi_1} + i_1) = 0$

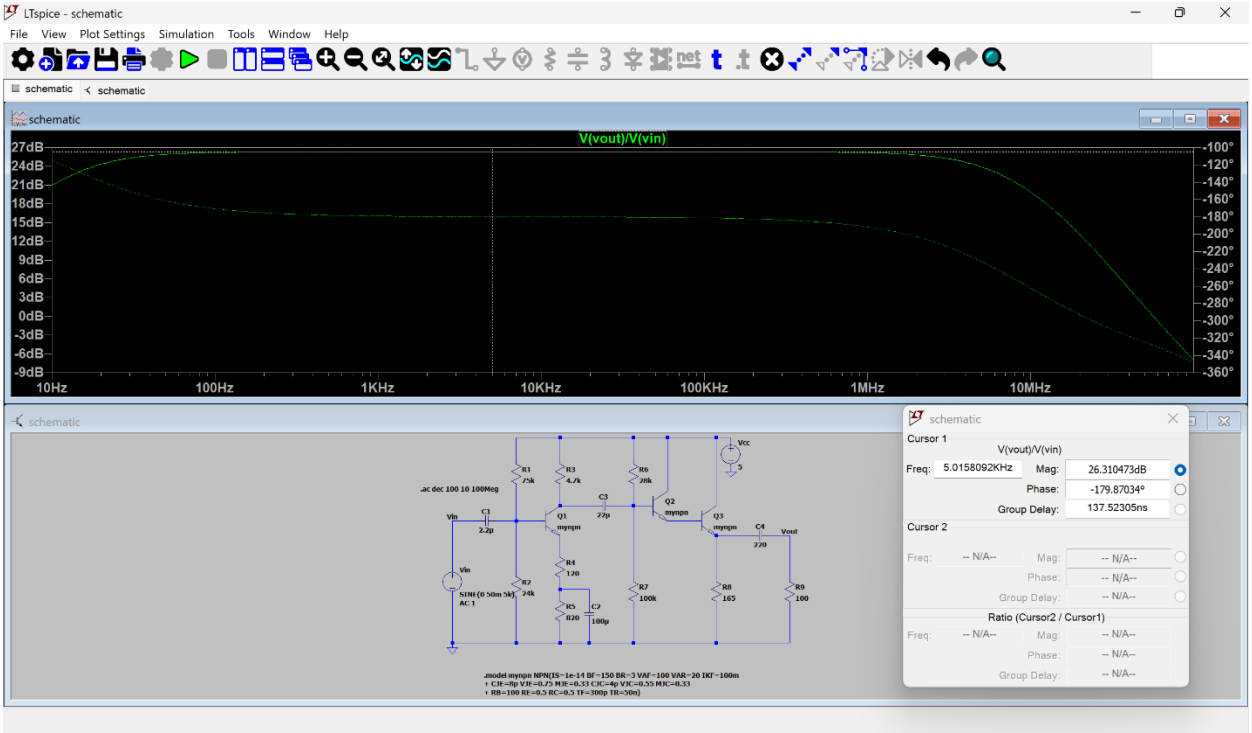
$\rightarrow i_1 \simeq -0.45\,v_{\pi_1}$

$21.74v_{\pi_1} - 0.45v_{\pi_1} = 21.29v_{\pi_1}$

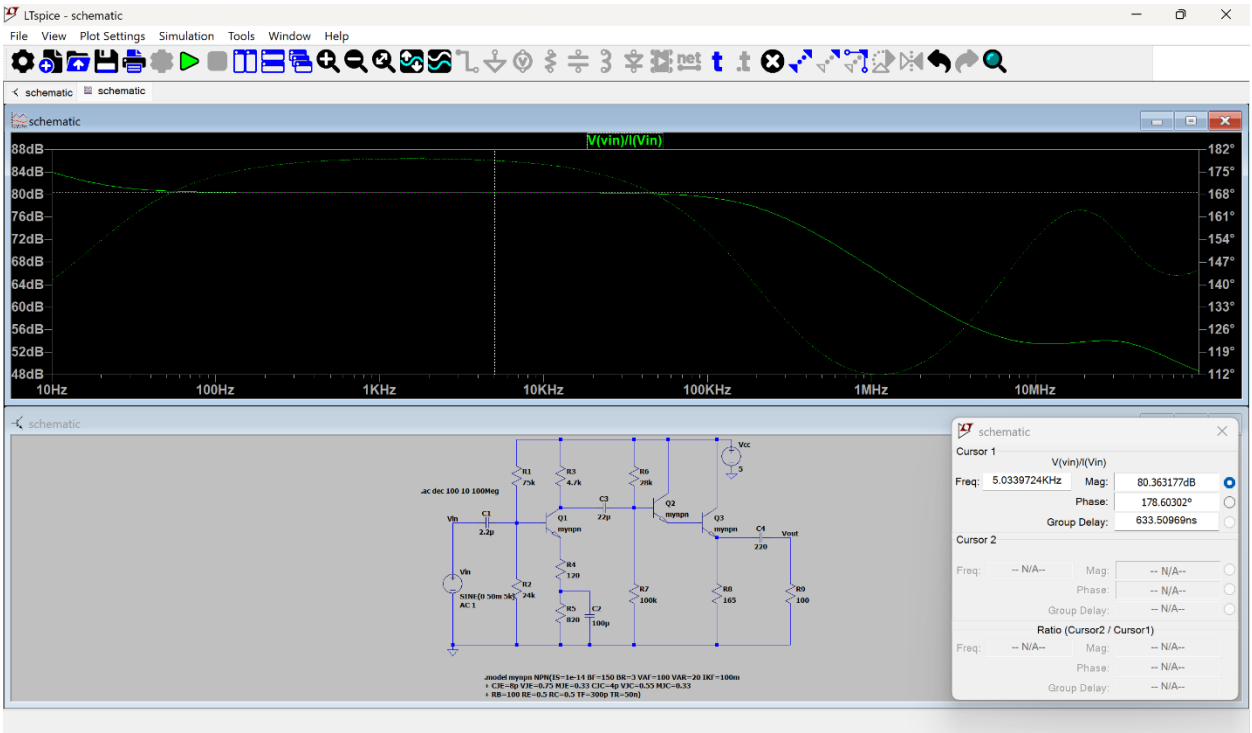
$V_{in} = v_{\pi_1} + 0.12 \times 21.29v_{\pi_1} \simeq 3.55\,v_{\pi_1}$

$i_T = 0.14v_{\pi_1} + \frac{3.55v_{\pi_1}}{18.18} \simeq 0.34\,v_{\pi_1}$

$R_{in} = \frac{3.55v_{\pi_1}}{0.34v_{\pi_1}} \simeq 10.44\,k\Omega$



مقدار بهره ولتاژ: 20.7



مقاومت داخلی: 10.4 k

.3

Swing:

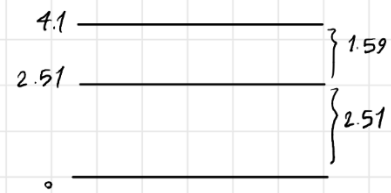
$5 - V_{CEsat} = 5 - 0.2 = 4.8 = V_{E2max} = V_{B3max}$

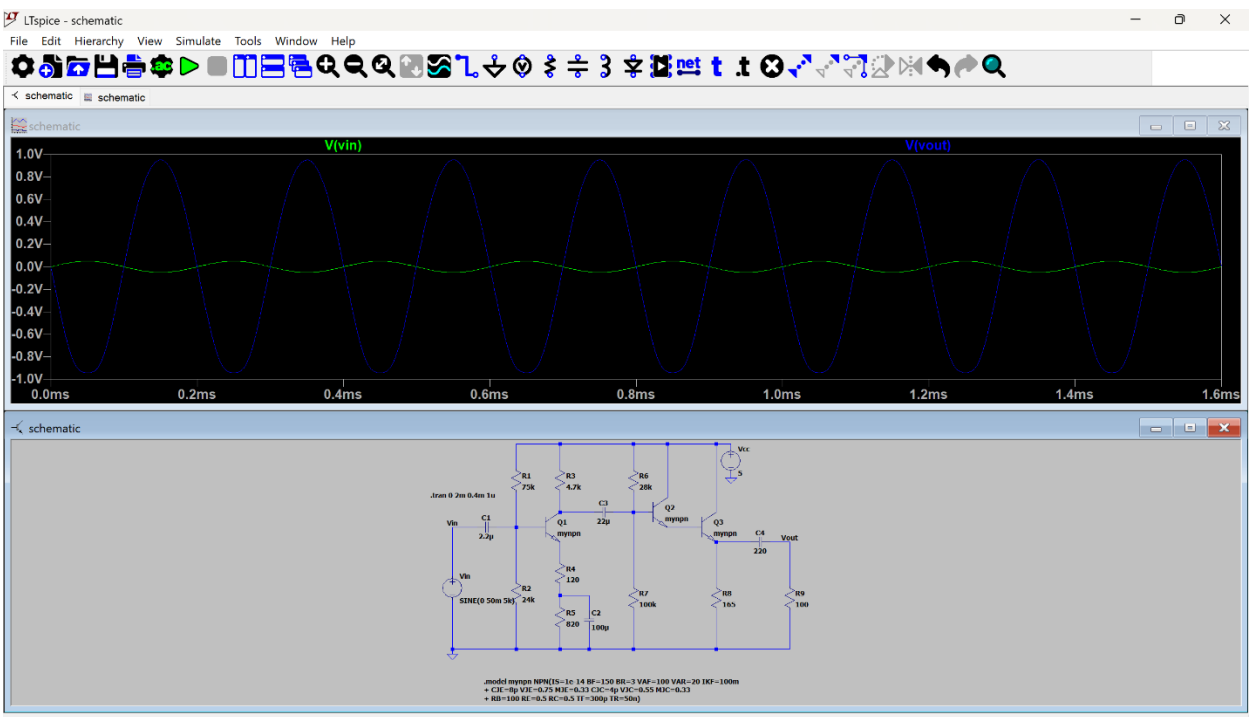
$V_{out\,max} = 4.8 - 0.7 = 4.1$

$V_{out\,min} = 5 - 5 = 0$

خط کار $V_{E3} = 2.51$

حداکثر دامنه خروجی قبل از برش ← 1.59 V

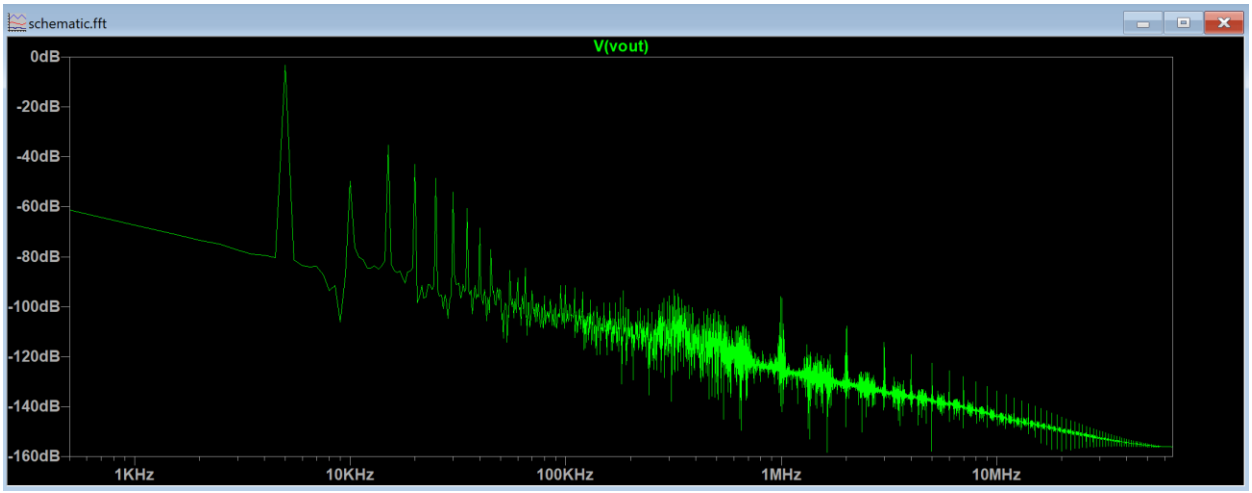




آیا سیگنال دچار کلیپینگ شده است؟

خیر زیرا با این ورودی، حداکثر خروجی 1 ولت است که از حداکثر دامنه خروجی قبل از برش کمتر است.

.5



SPICE Output Log: C:\Users\sohan\OneDrive\Desktop\university\el...

```
LTspice 24.1.5 for Windows
Circuit: C:\Users\sohan\OneDrive\Desktop\university\e
Start Time: Fri Feb 13 17:20:25 2026
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.179 seconds.

Files loaded:
C:\Users\sohan\OneDrive\Desktop\university\electronic
C:\Users\sohan\AppData\Local\LTspice\lib\cmp\standarc

Fourier components of V(vout)
N-Period=1
DC component:0.000649373

Harmonic     Frequency     Fourier     Normalized    Phas
 Number        [Hz]       Component    Component   [degr
    1        5.000e+03    9.800e-01    1.000e+00     -90
    2        1.000e+04    5.055e-03    5.158e-03      -0
    3        1.500e+04    2.386e-02    2.435e-02     -90
    4        2.000e+04    9.900e-03    1.010e-02      -1
    5        2.500e+04    5.258e-03    5.365e-03      88
    6        3.000e+04    2.682e-03    2.737e-03    -179
    7        3.500e+04    1.361e-03    1.389e-03     -90
    8        4.000e+04    5.098e-04    5.202e-04      -3
    9        4.500e+04    1.572e-04    1.604e-04      98
Partial Harmonic Distortion: 2.756715%
Total Harmonic Distortion:   2.757054%
```

مقدار THD نهایی:

2.76 درصد